Predicting MLB Players Homeruns
Using Last Seasons Stats.

Data: 'https://github.com/coltonmurph/IS-327-ML-Project/blob/main/MLBSavantStats200.csv'
This Dataset is from Baseball Savant which allows me to choose the features I want. It contains player statistics from the years 2017 -2026 that have at least 200 plate appearances. It columns include a Player ID, Player name, Year, plate appearances, Home Runs, K%, BB% ISO, xSLG, xwOBA, Avg Ev(MPH), AVG LA, LA Sweet-Spot%, Barrel%, Hard Hit%, Pull%, FB%

I'll match each player's seasons with his next season using player ID. I only use seasons with at least 200 plate appearances. We will use it to predict home runs per 600 plate appearances

I intend to apply **multiple linear regression** using my many features to predict since my number is not a category and most of my categories are mostly linear.

I will use **random forest** as well since some of my relationships aren't linear, and will allow me to understand which features are most important.

Research Question: Which Features best predict a hitters home runs per 600 plate appearances next season, traditional stats or statcast metrics? Does combining them improve predictions?

Hypotheses:

1. Statcast Metrics will predict a player's HR per 600 plate appearances better than last season's home run total because HR's include luck and barrel rate.
2. The combine feature set will perform the best
3. Random Forest will beat linear regression

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From doing this project, I hope to learn what is the most important indicator for predicting home runs, and gain an edge next year in my yearly homerun prediction contest by using this model.